

Case Study

Contents

■ Background	8-2
■ The Business Problem	8-3
■ Exercise 8–1: Your Assignment	8-5
■ Invoke Business Process Service Per Customer	8-9
■ The File System Adapter Process	8-14
■ The Command Line 2 Adapter Process	8-16
■ Lesson Review	8-18

Background

Introduction

This lesson provides an opportunity to analyze a business problem and create a business process based on the problem.

Lesson objectives

This lesson is designed to help you to:

- Analyze a business problem
 - Create a business process to solve the business problem
-

The Business Problem

Introduction

The Company agrees to perform a Vendor Managed Inventory service for two customers, My Little Store (MLS) and The Other Guy (TOG), both retailers are merchandise for the company.

The company auditor, requested to maintain a copy of the original data, so we must move a copy of the file to an archive directory. All processing should be automated without manual handling of data. It is already communicated and the data file(s) are now placed in /home/student/fscoll.

This exercise helps you understand more about bootstrapping, creating rules, XPath statements, and configuring different Services.

There are a number of different solutions that you can use to solve this case study, but one is given in detailed steps in this lesson.

The example solution that is presented in the detailed steps of this case study will have you build bootstrapping on a FSA into a business process instead of scheduling a FSA configuration like we did in the previous lesson. Both solutions work but scheduling is more commonly used in the real world. This solution does show the flexibility that is available with business processes and services.

Important!	When using bootstrapping or invoke, the business process that is invoked should be built, saved, and checked-in before it is referenced or used. This affects the order that Processes may be built, as you see in the following exercise.
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Important!	These exercises will have you make changes to service configurations in the GPM. Make sure to check Override Service Configuration under Options -> Preferences -> Service Editor. Please note that when overriding service configurations the overrides will only affect the bp where they are made. They will not globally affect other bp's that use that configuration.
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The Business Problem

(Continued)

Introduction (Continued)

In this solution, the following four different Business Processes need to be created:

1. **The Main Business Process:** Here build a business process that determines the Customer Id and take one of two paths, path for “My Little Store” or path for “The Other Guy”. This process invokes another Business Process that will determine whether it requires to order more items. After the business process runs, write out the file into the directory **/home/student/fs_output**. Suggested name: Main.BP.
 2. **The Invoke Business Process Service:** Here build two different Customer business processes to determine whether it needs to order more items. One business process for the “My Little Store” and another for “The Other Guy”. The order level for My Little Store is less than 10, then it should send more items. The order level for The Other Guy less than 20. Suggested names: LittleStore.BP and OtherGuy.BP
 3. **The Command Line Adapter Process:** Here build a business process that copies data into the correct folders, by calling the script, copytest.bat. This script looks for files in /home/student/fscoll with a name of *.xml. Then, move one copy of the files to /home/student/fsprocess and one copy to /home/student/fsarchive.
 4. **The File System Adapter Process:** Here build a process to PICK UP the files from /home/student/fsprocess and cause the Main.BP process to start running by using bootstrapping.
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Exercise 8–1 Your Assignment

Instructions	For ease of understanding, this procedure is covered in three parts.
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- Setup Steps
- The Main Business Process
- Procedure for the Main Process

Setup Steps	■ Verify whether the following directories were created: ■ /home/student/fsarchive ■ /home/student/fsprocess ■ /home/student/fsoutput ■ Verify that script copytest.sh is in the directory /home/student.
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The Main Business Process	This process determines the Customer ID and then takes a path for “My Little Store” or “The Other Guy”. Then invoke a Business Process Service that determines whether it is required to order more items. At the end of the process, write file(s) into the directory /home/student/fs_output. Suggestion: Build LittleStore.BP and OtherGuy.BP first (go to Invoke Business Process Service / Per Customer), then come back and build this process.
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Exercise 8–1 Your Assignment

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The Main Business Process (Continued)

In this process, the following stencils are required.:

Start (1) Choice Start (1)

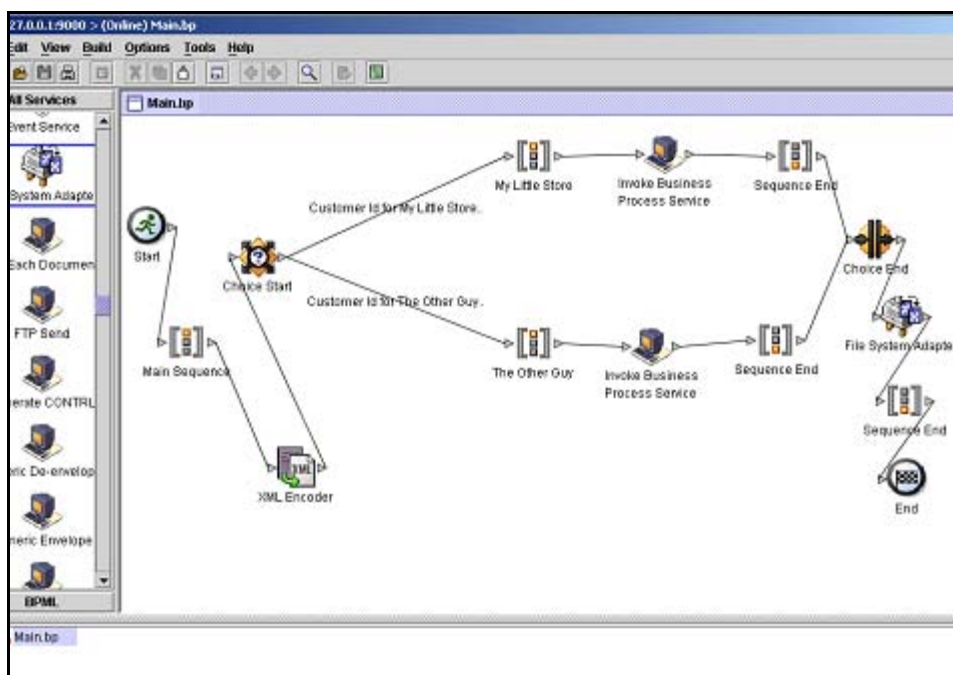
End (1) Choice End (1)

Sequence Start (3) XML Encoder Service (1)

Sequence End (3) File System Adapter (1)

Invoke Business Process (2)

The following figure depicts an example process.



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Exercise 8–1 Your Assignment

(Continued)

Procedure for the Main Process

You can follow the mentioned procedure for the main process.

1. Drag and connect your Stencils.
2. Changing the name of the 3 sequence starts.
 - a. Double-click the first sequence start and type in “Main Sequence” under the value column.
 - b. Double-click the top sequence start and type in “My Little Store” under the value column.
 - c. Double-click the bottom sequence start and type in “The Other Guy” under the value column.
3. Configure the XML Encoder:
 - a. Mode = Use existing XML document.
 - b. Output to process data = Yes.
4. Click **Tools > Rules Manager**. Click **Add** to bring up the Rule Editor. Populate the name box with “**Check for MLS**”. In the expression box, type the following Xpath string:

```
//INVENTORY/CUST_ID/text()='My Little Store'
```

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Exercise 8–1 Your Assignment

(Continued)

Procedure for the Main Process (Continued)

5. Click **Tools > Rules Manager**. Click **Add** to bring up the Rule Editor. Populate the name box with “**Check for TOG**”. In the expression box, type the following Xpath string:

```
//INVENTORY/CUST_ID/text()='The Other Guy'
```
6. Double click the line between **Choice Start** and the top **Sequence Start**. Click **Add** to open the name and value columns in the top Edge Editor. Select “**Check for MLS**” for the name and **true** under the value column.
7. The Invoke Business Process Service.
 - a. Invoke_mode = Sync
 - b. WFD_Name = LittleStore_#.BP or (whatever you named the business from the below exercise).
8. Double click the line between the **Choice Start** and the bottom **Sequence Start**. Click **Add** to open the name and value columns in the bottom Edge Editor. Select “**Check for TOG**” for the name and **true** under the value column.
9. The Invoke Business Process Service.
 - a. Invoke_mode = Sync
 - b. WFD_Name = OtherGuy.BP or (whatever you named the business from the below exercise).
10. The File System Adapter Service.
 - a. Action = Extraction.
 - b. AssignFilename = Use the original filename
 - c. Bootstrap = No
 - d. ExtractionFolder = /home/student/fsoutput
11. Save as Main.BP, validate and check in the business process.

Congratulations you can now add business process building to your Resume, but you are not done yet. Continue with the File System Adapter Process.

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Invoke Business Process Service Per Customer

Overview These two Customer business processes, My Little Store (MLS) and The Other Guy (TOG), will be invoked by the Main business process. It is required to build two copies of business process, one for “My Little Store” and another for “The Other Guy”. Build one, save, and validate; then do a Save As to create the second one with the other name.

In this process, the following stencils are required:

Start (1) Choice Start (1)

End (1) Choice End (1)

Sequence Start (3) Assign (2)

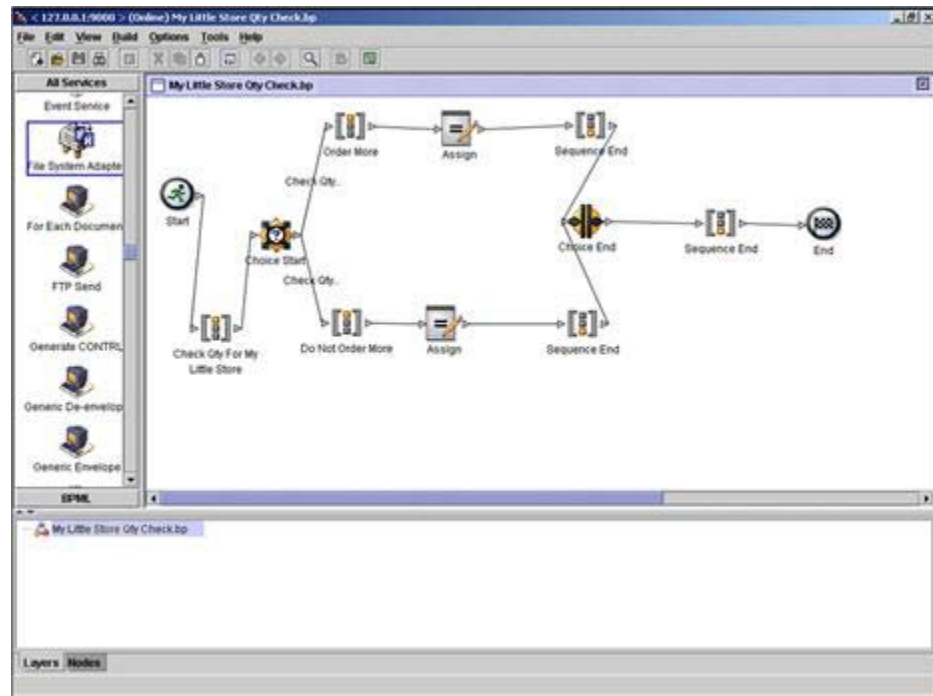
Sequence End (3)

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Invoke Business Process Service Per Customer (Continued)

Overview (Continued)

The following figure is an example of the process:



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Invoke Business Process Service Per Customer (Continued)

Step 1 for Customer Process

You can invoke business process per customer by following the steps that are listed:

1. Drag and connect your Stencils.
2. Changing the name of the 3 sequence starts.
 - a. Double-click the first sequence start and type in “Check QTY for My Little Store” in the value column.
 - b. Double-click the top sequence start and type in “Order More” under the value column.
 - c. Double-click the bottom sequence start and type in “Do Not Order More” under the value column.
3. Click **Tools > Rules Manager**. Click **Add** to bring up the Rule Editor. Populate the name box with “**Check QTY**”. In the expression box, type the following Xpath string:

```
sum(INVENTORY/PRODUCT/QTY) > 10
```
4. Click **Add** to open the name and value columns in the top Edge Editor. Select “**Check QTY**” for the name and **not true** under the value column.

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Invoke Business Process Service Per Customer (Continued)

Step 1 for Customer Process (Continued)

5. Click the **Assign** activity in Order More sequence and following information.

Name	Value
Constant	Order More Items
To	Results

6. Click **Add** to open the name and value columns in the bottom Edge Editor. Select "**Check QTY**" for the name and **true** under the value column.
7. Click the **Assign** activity in Do Not Order More sequence and following information.

This creates TAG in the output file: <results> Do not order more items</results>.

Name	Value
Constant	Do Not Order More Items
To	Results

8. Save as LittleStore.BP, validate and check in the business process.

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Invoke Business Process Service Per Customer (Continued)

Step 1 for Customer Process (Continued)

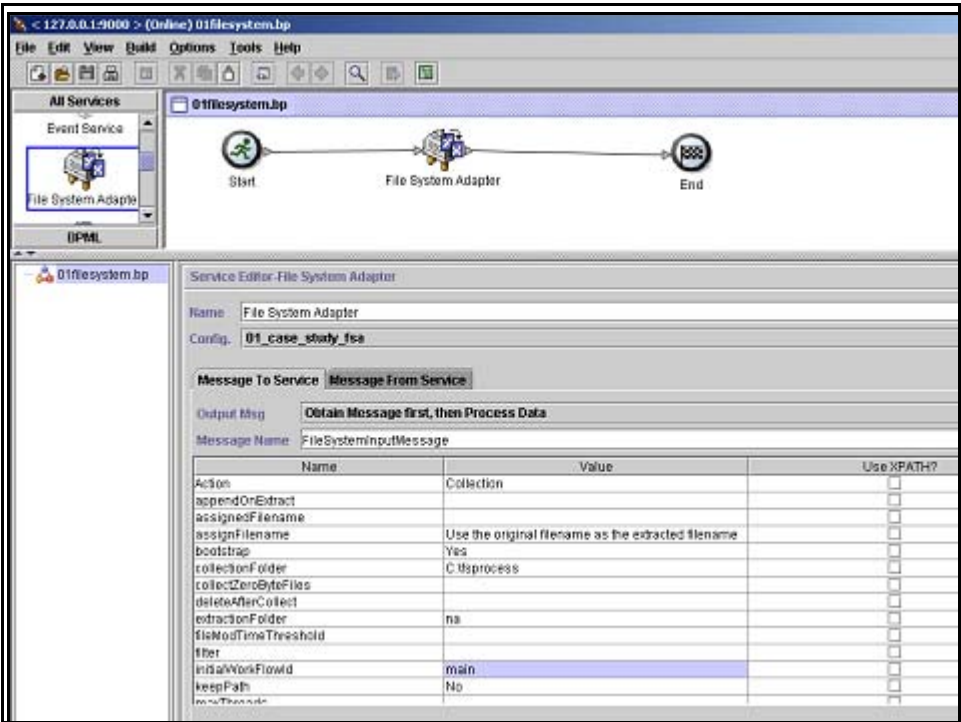
Perform a **Save As** to create a second business process for “The Other Guy”. Remember that the Check QTY is different for this customer, (20) and the starting sequence should be named “Check QTY for The Other Guy”.

1. Save as **OtherGuy.BP**, validate, and check in the business process.
 2. Now the Main Process.
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The File System Adapter Process

- Overview

This process picks up the files from /home/student/fsprocess and then starts the Main business process. In this process, the following stencils are required:
Start (1)
End (1)
File System Adapter (1)
The following figure depicts an example process.



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The File System Adapter Process

(Continued)

Overview (Continued)

3. Drag and connect your Stencils.
 4. Open the Service Editor for the File System Adapter Service.
 - a. Action = Collection
 - b. Assign Filename = Use the original filename.
 - c. Bootstrap = Yes
 - d. CollectionFolder = /home/student/fsprocess
 - e. DeleteAfterCollect = Yes
 - f. InitialWorkFlowid = main.bp
 - g. FileModTimeThreshold = 0
 5. Save as **KICKOFF.BP**, validate and check in the business process.
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The Command Line 2 Adapter Process

Overview

This process starts the business flow by calling a script that is called, copytest.sh. This script is set to look for files in /home/student/home/student//fscoll with a name of inventoryof3a.xml. After picking up the files, the script will make two copies of the file, depositing one file in /home/student/fsachive and the other in /home/student/fsprocess and then starting the File System Adaptor business process. (the previous process)

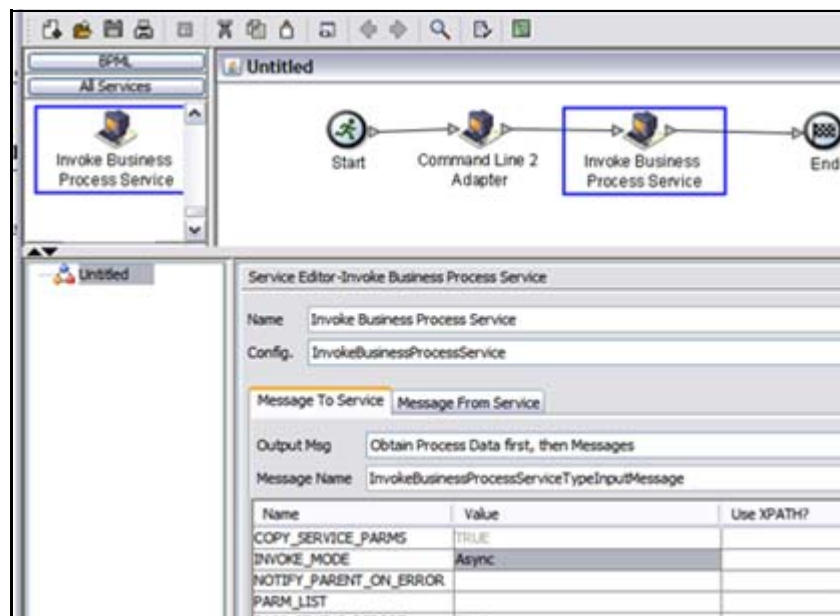
In this process, the following stencils are required:

Start (1)

End (1)

Command Line 2 Adapter (1)

The following figure depicts an example process.



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The Command Line 2 Adapter Process (Continued)

Overview (Continued)

1. Drag and connect your Stencils.
2. Open the Service Editor for the Command Line 2 Adapter Service.
 - a. Bootstrap = No
 - b. CmdLine = /home/student/copytest.sh
 - c. WorkingDirectory = /home/student.
 - d. InputDelete = true
 - e. OutputDelete = No
 - f. WaitOnProcess = Yes
3. Add an Invoke Business Process stencil between the command line 2 Adapter and the end stencil. Set the following parameters.
 - a. WFD_NAME = KICKOFF.bp (the bp from the previous step)
 - b. INVOKE_MODE = Asynchronous
4. Save as START.bp, validate, and check in the business process.

You can modify existing Service Configurations, or create new Service Configurations for use in this case study.
5. Locate the data files in the Case Study Labfiles folder under the 915 Fundamentals Folder and copy/paste all of the MLS and TOG files into the collection folder (/home/student/fscoll).

When you cause START to run manually, copytest.sh copies the XML files to the fsarchive directory and the fsprocess directory; then it uses the Invoke Business Process to start Kickoff. Kickoff pulls in the files from fsprocess and starts the Main Business Process.

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Lesson Review

Completed Objectives

This lesson was designed to help you to:

- Analyze a business problem
 - Create a business process to solve the business problem
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